

HARSH SAW

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SUMMARY

Full-stack software engineer with 2+ years of experience shipping production systems across distributed messaging, cloud infrastructure, and AI pipelines. Currently building RAG and agentic workflows at Ommuse; prior work spans real-time marketing platforms, video streaming infrastructure, and mobile apps serving thousands of users.

EDUCATION

Bachelor of Computer Information Systems, Okanagan College

Expected Dec 2026

EXPERIENCE

Software Engineer, AI & Automation | [Okanagan College — IT Services](#), Kelowna, BC

Apr 2026 – Present

- Building OC Connect, a college-wide unified-services platform — an agentic AI assistant on Azure AI Foundry that goes beyond retrieval to take action for staff and students, surfacing recommendations and initiating cross-system requests.
- Architected a custom RAG layer over heterogeneous institutional data, with a microservice fabric integrating Banner SIS and multiple platform APIs behind a unified retrieval and action interface.
- Shipped an org-wide workflow and issue-tracking system adopted across departments — Jira-style process tracking with real-time collaboration, replacing manual coordination between staff, instructors, and students.

Software Engineer | [Ommuse](#), Seattle, US

Nov 2025 – Present

- Architecting **AI-powered semantic search** and natural language chat for enterprise studio workflows using a **RAG** pipeline (**MongoDB Vector Store**, **AWS Bedrock**), enabling producers to query projects, tracks, and collaborators conversationally.
- Built **HubSpot CRM** sync for a 12K+ user music platform in **Go** — async batch worker using **sync.Map** deduplication (30s flush, 100-user chunks) syncing live metrics across 8 API trigger points; platform manages 18K+ tracks and 67 GB of user audio.
- Engineered a cross-platform desktop uploader (**Electron**) with persistent folder-sync engine (**chokidar** + **SQLite WAL**), parallelized **S3 multipart uploads**, **OAuth 2.0 PKCE** auth flow, and a typed IPC bridge exposing 70+ **gRPC** service methods to the renderer.
- Shipped features across a polyglot monorepo (**Go gRPC/protobuf**, TypeScript/React, Python) unified by **Bazel**, owning full deployment infrastructure — **AWS** (S3, SQS, ECR), **Docker**, **Pulumi** IaC, Envoy proxy, and **GitHub Actions** CI/CD.

Software Engineer Intern | [Bwisher Ltd](#), Remote

Dec 2024 – Jun 2025

- Architected a multi-channel marketing platform using **Apache Kafka** and **BullMQ**, orchestrating personalized SMS, email, and **WhatsApp** campaigns with custom offers across **30K+** users while feeding real-time CRM and analytics pipelines.
- Designed idempotent **NestJS** and **FastAPI** microservices for payment and credit workflows with retry and exponential backoff strategies, eliminating duplicate transactions in high-stakes financial operations.
- Owned end-to-end deployment infrastructure with **Jenkins**, **Kubernetes**, **Prometheus**, and **Grafana**; shipped features behind feature flags enabling safe tenant-by-tenant rollouts with instant kill-switch control.

Full-Stack Developer Intern | [Jythu Ltd](#), Remote

Jan 2024 – Dec 2024

- Shipped two production **React Native** apps (LMS + admin panel) with **Node.js** backends serving **2K+ learners**; provisioned full **AWS** infrastructure with **Terraform** and **GitHub Actions** CI/CD, with multi-layer anti-piracy controls including device-bound auth.
- Built an end-to-end video streaming pipeline using **FFmpeg** HLS transcoding delivered via **S3/CloudFront**, orchestrated through **SQS**, **Lambda**, and **SNS** — reducing seek time from **30s to 1.5s** and cutting bandwidth by approximately **70%**.

PROJECTS

Document Intelligence Platform (RAG) *FastAPI, LangChain, Qdrant, Groq, Google Gemini, AWS ECS Fargate, Docker*

[Live Demo](#) — [GitHub](#)

- Architected a multi-tenant RAG platform with session-isolated **Qdrant** indexes, enabling semantic search and conversational Q&A over PDF, DOCX, and TXT documents.
- Engineered resilient LLM orchestration using **LangChain RunnableWithFallbacks**, implementing automatic provider failover (Groq to Gemini) with streaming retrieval pipelines featuring context-aware query rewriting and hybrid document chunking.
- Deployed to **AWS ECS Fargate** with GitHub Actions CI/CD and multi-worker Uvicorn scaling; designed idempotent ingestion using **SHA-256** fingerprinting to deduplicate embeddings and reduce redundant indexing.

CodeExpo – Browser-Based IDE & Sandbox Environment *React, Node.js, WebSocket, Docker, Nginx, Zustand, xterm.js*

[Live Demo](#) — [GitHub](#)

- Engineered a full-stack browser-based IDE replicating a local dev environment — live terminal via **xterm.js** and **WebSocket**, real-time code execution with browser preview pane, and **Socket.IO** file synchronization across the workspace.
- Architected containerized sandbox environments using **Docker** with multi-stage builds and **Nginx** reverse proxy, managing isolated execution contexts and environment lifecycle per session.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Go (Golang), Bash

Frontend & Mobile: React, Next.js, React Native, Qwik.js, Electron

Backend & Messaging: NestJS, FastAPI, Express, Fastify, GraphQL, gRPC/Protobuf, Apache Kafka, BullMQ, RabbitMQ

Databases: PostgreSQL, MongoDB, DynamoDB, Redis, MySQL, SurrealDB, SQLite

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, Pulumi, Bazel, Jenkins, GitHub Actions

AI & ML: PyTorch, Hugging Face Transformers, PEFT/LoRA/QLoRA, Unsloth, vLLM, embeddings, cross-encoder reranking, LangChain, RAG, Qdrant, FAISS, Pinecone, AWS Bedrock

ACHIEVEMENTS

- Sole developer of an internal LMS migration platform at **Okanagan College**, built and deployed end-to-end and relied on daily by **1,500+ staff and instructors** for the institution-wide Moodle-to-Brightspace transition — featuring guided step-by-step migration workflows with progress tracking and complete records, real-time notifications and chat, and AI-powered templating.
- Built an end-to-end stock prediction pipeline covering **500 S&P 500 constituents** over 5 years of 15-minute bar data — **XGBoost** models trained across 26 intraday horizons on **4x NVIDIA H100 GPUs** (DRAC), with automated warm-start retraining every 15 minutes during market hours, served via a **FastAPI** backend backed by a **PostgreSQL** data warehouse and a multi-rolling feature pipeline.
- Ranked in the **top 2% of verified freelancers** on [Freelancer.com](#) with a 5-star rating, delivering 25+ production-grade projects.
- Won an inter-university college hackathon, demonstrating innovative problem-solving and technical proficiency.